Electronically Filed Case P36373-EP
European Patent Office
80298 Munich
Germany

Basel, December 21, 2023 SMU/LG

Applicant: Genentech Inc.
Application No.: 21824724.5 - 1118 Case: P36373-EP

Dear Sir/Madam,

Reference is made to the communication pursuant to Rules 161(1) and 162 EPC dated 29.06.2023.

Applicant herewith files a new set of claims 1-34 which replaces claims 1-36 as currently on file. A clean version and a marked version showing the amendments are enclosed.

It is respectfully requested to continue the examination on the basis of the newly submitted claims. The applicant reserves the right to pursue any presently unclaimed subject matter in the current proceedings or in one or more divisional applications pursuant to Article 76(1) EPC.

Amendments

Claim 1 has been amended by including dependent claims 6 and 7 specifying that the detection molecules are cross-linked to the matrix with a crosslinker selected from dimethylpimelimidate (DMP), a cyanate ester, an NHS ester, azlactone, carbonyl diimidazole (CDI), maleimide, iodoacetyl, pyridyl disulfide, hydrazide, or a carbodiimide. Basis for this amendments can be found for example in claims 1, 6 and 7 as originally filed.

Claims 13 to 15 have been included as alternatives in claim 12. Basis for this amendment can be found in the claims as originally filed.

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The dependencies and claim numbering has been adjusted accordingly.

It is respectfully requested to postpone adaptation of the description until the claims are allowable.

No new subject-matter has been added. Accordingly the new claims meet the requirements of Article 123(2) EPC.

Novelty A. 54(2)

The applicant agrees with the Examiner that claims 4, 7, 9, 16, 17, 19, 20, 22-26, 30, 32-34 and 36 are novel. As claim 7 has been included in claim 1, all claims as currently on file are also novel.

Applicant believes that the new set of claims filed herewith comply with the requirements of Article 54 EPC.

Inventive step A. 56

The Examiner alleges that none of the claims are inventive. The applicant respectfully disagrees. D1 allegedly discloses a cartridge containing magnetic particles coated with antibodies and sucrose. D2 allegedly discloses a cartridge for detecting an analyte in solution comprising agarose beads bound with monoclonal antibodies and comprising glycerol.

D1 or D2 can be regarded as the closest prior art. The difference between the present invention and the closest prior art is that the detection molecules are cross-linked to the matrix with a cross-linker selected from dimethylpimelimidate (DMP), a cyanate ester, an NHS ester, azlactone, carbonyl diimidazole (CDI), maleimide, iodoacetyl, pyridyl disulfide, hydrazide, or a carbodiimide.

The effect of this difference is that detection molecules are prevented from being leached from the matrix over repeated use as for example stated in paragraph [0054] of the application as originally filed.

The problem can be formulated as how to provide reusable cartridges for detecting an analyte in solution. D3 relates to nano-ultra-performance liquid chromatography coupled to high-resolution mass spectrometry for characterization of peptides bound to MHC molecules, while the present application relates to cartridges which can be used before chromatography and mass spectrometry (see paragraph [0057] and [0058] of application as originally filed). D3 thus relates to a different field. D3 does disclose DMP as a cross-linker in a different field without suggesting that any material might be reused.

None of the prior art discloses or suggests to use the claimed cross linkers to prevent detection molecules from being leached from the cartridge over repeated use thereby providing a solution to the formulated problem.

For the reasons mentioned above, applicant believes that claims filed herewith comply with the requirements of Article 56 EPC.

Conclusion and Requests

It is believed that the currently claimed subject matter is novel and inventive as explained above.

The Examining Division is respectfully requested to continue the examination of the present application and to envisage granting the patent sought for. In case further objections arise, a first communication according to Art. 94(3) EPC is requested. If the Examining Division decides to refuse this application without issuance of a communication according to Art. 94(3) EPC, oral proceedings (Art. 116 EPC) are herewith requested.

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Enclosures: mentioned